UPI Transaction Fraud Detection Analytics – India

1. Business Context

India has experienced explosive growth in digital payments through the Unified Payments Interface (UPI).
Millions of transactions occur every minute across banks, mobile devices, and network infrastructures.

With the rapid adoption of UPI, financial institutions and payment service providers face increasing challenges such as:
- Fraudulent peer-to-peer transfers
- Suspicious merchant payments
- Identity misuse through compromised devices
- Transaction anomalies across states and banking networks

Even a small fraud rate can result in millions of rupees in financial losses and reputational damage.
Banks and payment platforms require data-driven fraud monitoring systems capable of identifying suspicious
transactions before or immediately after they occur.

2. Problem Statement

A financial risk analytics team has been provided with a dataset containing 250,000 UPI transactions
across multiple Indian states, banks, devices, and transaction types.

The objective is to analyze transaction behavior and identify patterns associated with fraudulent activity.

Using transaction attributes such as transaction amount, merchant category, sender and receiver age group,
device type, network type, sender bank and receiver bank, geographic state, time of transaction, and day of week,
the goal is to determine:

1. Which behavioral patterns are most associated with fraud
2. Which regions or banks experience higher fraud risk
3. What operational signals could be used for early fraud detection

These insights will help financial institutions detect fraud earlier, reduce financial loss, strengthen risk monitoring frameworks, and improve transaction security systems.

3. Dataset Overview

The dataset contains 250,000 synthetic UPI transaction records with attributes including:

- transaction_id: Unique transaction identifier
- timestamp: Date and time of transaction
- transaction_type: P2P, P2M, Recharge, Bill Payment
- merchant_category: Spending category (Food, Grocery, Utilities, etc.)
- amount: Transaction value (INR)
- transaction_status: Success or Failed
- sender_age_group: Age group of sender
- receiver_age_group: Age group of receiver
- sender_state: State where transaction originated
- sender_bank: Bank initiating payment
- receiver_bank: Bank receiving payment
- device_type: Android, iOS, Web
- network_type: 3G, 4G, 5G, WiFi
- fraud_flag: Fraud indicator (1 = Fraud, 0 = Legitimate)
- hour_of_day: Hour transaction occurred
- day_of_week: Day name
- is_weekend: Weekend indicator

4. Business Objectives

The analysis aims to answer key fraud risk questions:

Fraud Risk Identification
Identify patterns and characteristics that differentiate fraudulent vs legitimate transactions.

Regional Risk Monitoring
Determine which Indian states show higher fraud activity.

Behavioral Fraud Analysis
Understand how fraud correlates with transaction amount, transaction timing, device usage, network type, and bank combinations.

Operational Fraud Indicators
Identify signals that can help build rule-based fraud detection alerts.

5. Key Analytical Questions (Scenario Based)

Transaction Behaviour Analysis
- What percentage of total transactions are fraudulent?
- Are fraudulent transactions associated with higher transaction amounts?
- Which transaction types experience the most fraud?

Geographic Risk Analysis
- Which Indian states show the highest fraud rates?
- Are there regional clusters of fraud activity?

Banking Network Analysis
- Which bank combinations (sender to receiver) show higher fraud occurrence?
- Do certain banks appear more frequently in fraud cases?

Device and Network Risk
- Are fraud transactions more common on Android, iOS, or Web devices?
- Does network type influence fraud likelihood?

Time-Based Fraud Patterns
- At what hours of the day do fraud transactions occur most often?
- Are fraud attempts more common on weekends versus weekdays?

Demographic Insights
- Which age groups are more frequently involved in fraud transactions?
- Are certain sender-receiver age group combinations suspicious?

6. Analytical Workflow

1. Data Understanding – Understand transaction attributes and fraud labeling.
2. Data Cleaning – Remove duplicates, validate timestamps, handle missing values, standardize categories.
3. Feature Engineering – Create derived features such as high value transaction flags and time buckets.
4. Exploratory Data Analysis (EDA) – Analyze fraud distribution across geography, transaction types, and time.
5. Fraud Pattern Detection – Identify behavioral anomalies and suspicious combinations.
6. Dashboard Development – Build an interactive fraud monitoring dashboard using Power BI.

7. Expected Deliverables

Fraud Risk Dashboard including:
- Total Transactions
- Fraud Rate
- Fraud Transaction Volume

- Average Fraud Amount

Visualizations:
- Fraud by State
- Fraud by Transaction Type
- Fraud by Device
- Fraud by Time of Day
- Fraud by Bank Network

8. Business Impact

Insights from this analysis can help payment providers detect fraud earlier, reduce financial losses, strengthen fraud monitoring rules, and improve trust in digital payment ecosystems.

9. Tools Used

Python – Data generation and preprocessing
SQL – Data querying and aggregation
Power BI – Fraud monitoring dashboard
Excel / Power Query – Data modeling
Pandas & NumPy – Data analysis

10. Project Use Case

This project simulates a real-world fraud analytics scenario in digital payments, demonstrating skills in financial transaction analysis, fraud detection analytics, large dataset exploration, and business intelligence reporting.